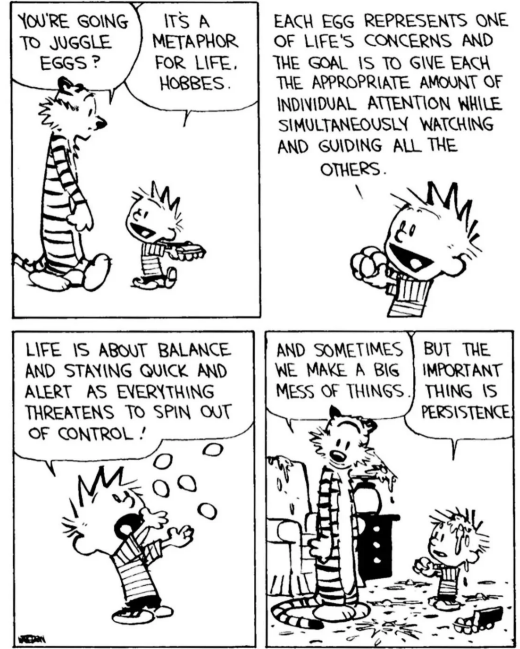




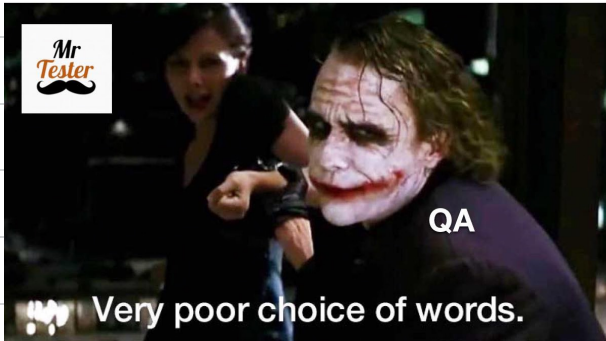
## AGENDA

- ① What is Docker??
- ② Diff b/w Docker, Virtual machine and virtual environments.
- ③ Workflow of Docker
- ④ Docker hub

# WHAT IS DOCKER

# WHY WE NEED IT?

Dev: Works in my machine...



Diff reasons this might  
be true?

- ① Library Versions
- ② Diff OS
- ③ Frontend  
backend  
Database  
ML



## Virtual Environments ??

Typically refers to  
Python packages

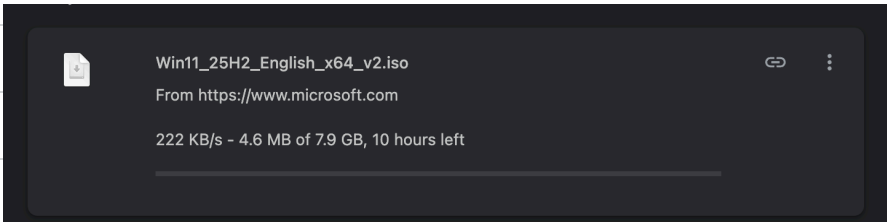
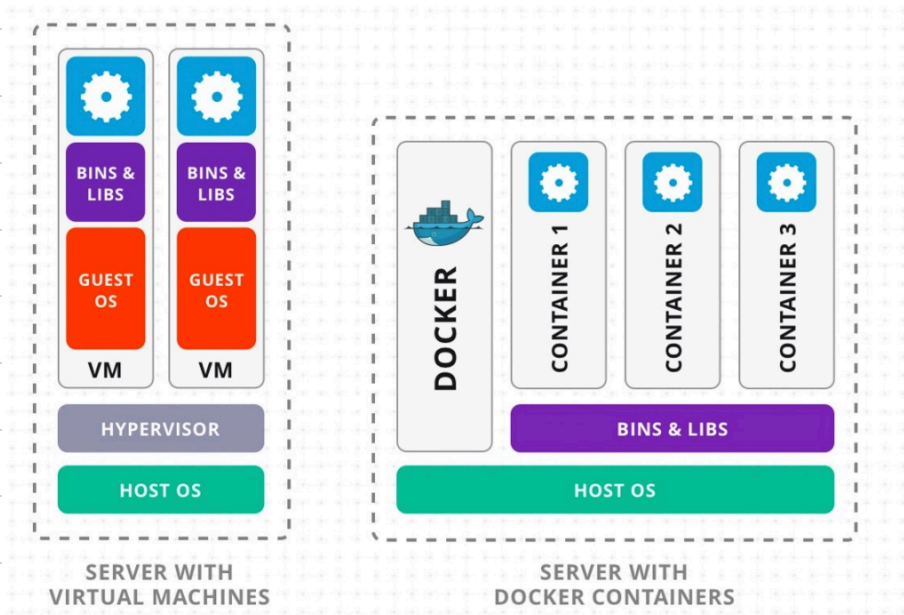
Docker



Isolated Environment / Container.

Mini - OS + Packages

Diff b/w Virtual Machines, Docker Container, and Virtual Environment.



64 GB RAM  
2TB SSD

① VM - windows

12 GB RAM, 50 GB SSD

40 GB RAM

② VM - Ubuntu

12 GB RAM, 50 GB SSD



# alpine

 Docker Official Image · [↓ 1B+](#) · [☆ 10K+](#)

A minimal Docker image based on Alpine Linux with a complete package index and only 5 MB in size!

OPERATING SYSTEMS

[Overview](#)

[Tags](#)

In every operating system, kernel is the lowest level of software in your OS, kernel interfaces directly with all your hardware and controls your applications.

Some responsibilities:

1. Process management and scheduling.
2. Memory management, like allocating RAM.
3. I/O handling
4. Supervising security.

Pros of Virtual Machine:

1. Virtual machines are extremely secure, isolated and they also give us exposure to diff os, like Kali Linux.
2. If you're working on something secretive/shady, you'll be fine.

Cons:

1. It is very very heavy weight.

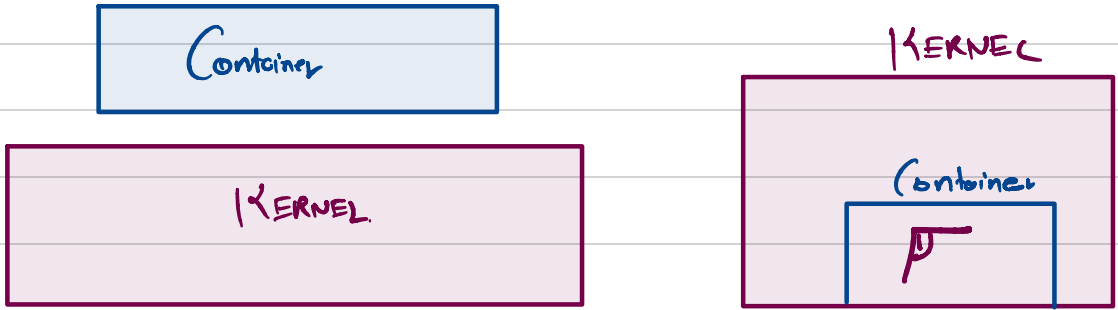
## What does containerization primarily achieve?

1 user has participated

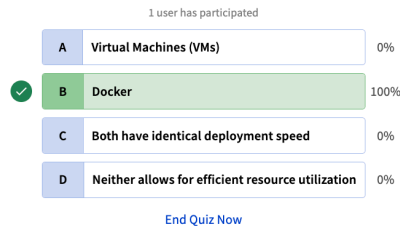
|                                                                                     |                                                                |      |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------|------|
| A                                                                                   | Optimization of graphical user interfaces                      | 0%   |
| B                                                                                   | Integration of hardware components                             | 0%   |
| C                                                                                   | Encryption of data transmission                                | 0%   |
|  D | Packaging software with necessary dependencies for portability | 100% |

[End Quiz Now](#)

|                                                                                     |                    |             |                                                                                     |                                                                                       |
|-------------------------------------------------------------------------------------|--------------------|-------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |                    |             |  |  |
| SAGAR GUPTA                                                                         |                    |             | Sanidhya Shukla                                                                     | ARIJIT SUR                                                                            |
| 1/1 ⚡ 96.15                                                                         |                    |             | 1/1 ⚡ 99.03                                                                         | 1/1 ⚡ 93.40                                                                           |
| 4                                                                                   | Rakesh Kumar Singh | 1/1 ⚡ 93.30 |                                                                                     |                                                                                       |
| 5                                                                                   | Manikandan A       | 1/1 ⚡ 93.16 |                                                                                     |                                                                                       |
| 6                                                                                   | Anirban            | 1/1 ⚡ 91.77 |                                                                                     |                                                                                       |
| 7                                                                                   | Nagaraj S H        | 1/1 ⚡ 91.63 |                                                                                     |                                                                                       |
| 8                                                                                   | J. Ramesh Reddy    | 1/1 ⚡ 88.87 |                                                                                     |                                                                                       |
| 9                                                                                   | JAWAJI RAGHUVEER   | 1/1 ⚡ 80.57 |                                                                                     |                                                                                       |
| 10                                                                                  | K. Janani          | 1/1 ⚡ 71.03 |                                                                                     |                                                                                       |



Which technology provides a more lightweight approach to virtualization, allowing for faster deployment and efficient resource utilization between Docker and Virtual Machines (VMs)?



|         |                    |                               |        |                           |  |
|---------|--------------------|-------------------------------|--------|---------------------------|--|
| AS<br>2 |                    | Sanidhya Shukla<br>2/2 188.20 |        | SAGAR GUPTA<br>2/2 185.27 |  |
| 4       | Anirban            | 2/2                           | 180.63 |                           |  |
| 5       | Manikandan A       | 2/2                           | 174.70 |                           |  |
| 6       | JAVVAJI RAGHUVIEER | 2/2                           | 161.70 |                           |  |
| 7       | Nageraj S H        | 2/2                           | 155.53 |                           |  |
| 8       | Rakesh Kumar Singh | 1/2                           | 93.30  |                           |  |
| 9       | Kuntal Kole        | 1/2                           | 92.43  |                           |  |
| 10      | J. Ramesh Reddy    | 1/2                           | 88.87  |                           |  |

## Deployment Steps

1. Find some suitable cloud environment (AWS, GCP, Azure, etc)
2. Install/Choose OS you want to work with -> Linux/Unix (Windows, MacOS)
3. Install dependencies and libraries (including front-end, back-end, DB, etc)
4. Copying the code from the repo to the server (current cloud server)
5. Run the code and expose the URL/API

## Docker Terminologies

1. DockerFile: Contains all the instructions, of all the packages/codes and any and all relevant files you need for your application (Recipe)
2. We use DockerFile to create "Docker Image" - Class
3. We use Docker Image to create Docker Container (instance of the class), Container runs the Docker Image.

In Docker, what does an "image" primarily represent?

0 users have participated

- A

A snapshot of the running containers

0%
- B

The underlying operating system of the host machine

0%
- C

A static, standalone package containing application code and dependencies

0%
- D

An encrypted file containing Docker configurations

0%

[End Quiz Now](#)

Based on all quizzes from the session



2



1



3

AS

 **SAGAR GUPTA**  
3/3 ⚡ 281.10

 **Sanidhya Shukla**  
3/3 ⚡ 296.23

 **ARIJIT SUR**  
3/3 ⚡ 278.53

|    |                                                                                                      |              |
|----|------------------------------------------------------------------------------------------------------|--------------|
| 4  |  Anirban            | 3/3 ⚡ 269.37 |
| 5  |  Manikandan A       | 3/3 ⚡ 264.03 |
| 6  |  JAVVAJI RAGHUVEER  | 3/3 ⚡ 226.13 |
| 7  |  Rakesh Kumar Singh | 2/3 ⚡ 184.60 |
| 8  |  J. Ramesh Reddy    | 2/3 ⚡ 178.47 |
| 9  |  K. Janani          | 2/3 ⚡ 162.73 |
| 10 |  Nagaraj S H        | 2/3 ⚡ 155.53 |



